

Vinay Pyatimani

Embedded Software Engineer

+49 176 732 43087 pyatimani.v@gmail.com

Karlsruhe, Germany

Nationality: Indian



EXPERIENCE

Marvell Semiconductors, Ettlingen, Germany Embedded Software Engineer

November 2018 - February 2025

- Designed and developed safety-critical firmware in C/C++ for Automotive Ethernet switches, covering the full software life cycle from requirements engineering through verification and maintenance.
- Led implementation of real-time automotive communication protocols: gPTP (IEEE 802.1AS) and MKA (IEEE 802.1X).
- Performed embedded Linux board bring-up and BSP integration for automotive hardware platforms; developed and maintained low-level device drivers.
- Performed deployment of firmware onto target embedded hardware; optimised performance using Lauterbach TRACE32 and perf profiling.
- Created modular, reusable firmware packages tailored to varying customer and system requirements.
- Ensured code quality and compliance via static analysis tools: PC-lint, Klocwork, and Polyspace; maintained MISRA-C compliance.
- Supported achievement of ASPICE L2/3 and maintained awareness of ISO 26262 functional safety requirements in all design and verification activities.
- Authored and reviewed unit tests and test specifications; drove test automation using GTest/CMock to improve verification coverage.
- Established and maintained CI/CD pipelines using Jenkins, Docker, and Ansible for automated build, integration, and deployment.
- Collaborated with cross-functional global teams (developers, testers, OEM contacts) across Germany and the US with a high degree of autonomy.

Infineon, Duisburg, Germany - Master Thesis

July 2017 - October 2018

- Developed a full embedded application on the Infineon XMC4800 microcontroller for an Industry 4.0 smart camera system, covering design, implementation, and verification.
- Implemented camera sensor interface and data pipeline, including driver-level integration between the image sensor and the communication stack.
- Developed an IP stack for data transmission using TCP/IP, MQTT, and OPC-UA protocols.
- Modified and optimised existing peripheral drivers to improve data throughput and system latency.
- Presented technical results to clients and external industry partners.

Mindtree Consulting, Bangalore, India Frontend Software Engineer

June 2011 – August 2015

- Frontend development using HTML, CSS, JavaScript, and jQuery for enterprise-grade web applications.

Job Related Abilities

Technical Knowhow

- Over 6 years of professional experience across the full software life cycle - requirements engineering, design, implementation, verification, and maintenance.
- Expert in modern C/C++ and embedded Linux platform development, with hands-on experience in time synchronization (PTP/gPTP), automotive communication protocols, sensor integration, and board bring-up.
- Proven track record in DevOps, CI/CD pipelines, and test automation.

Communication

- 4+ years of experience in cultivating, and managing communication channels between customers, clients, internal and external stakeholders
- Well-versed in identifying and tailoring both oral and written communications to specific learning and personality types in German and English languages

Technical Proficiency

- ANSI C, Advanced C++14/17/20, Python, Shell scripting (Bash/sh)
- Ethernet, TCP/IP, CAN, Automotive Ethernet (100BASE-T1/1000BASE-T1), EtherCAT, MQTT
- Time synchronisation: PTP (IEEE 1588), gPTP (IEEE 802.1AS), PPS
- Build systems: Makefile, GCC, Clang; Version control: Git
- Debugging: GDB, Lauterbach TRACE32, Segger, STLink(v2); Profiling: perf, Lauterbach
- CI/CD pipelines: Jenkins, Docker, Ansible
- Static analysis: PC-lint, Klocwork, Polyspace
- Project management: Polarion, DOORS, Jira

Leadership

- Mentoring student workers, assigning tasks, keeping track of their progress at Marvell.
- Lead a 3 people team for unit testing: looked over tasks, code coverage, test results and test specifications at Marvell.
- Lead a team of 4 junior developers at

- Translated customer requirements into software specifications; engaged directly with clients for design changes and feature prioritisation.
- Mentored junior engineers and managed a sub-team of 3 developers.

EDUCATION

Fachhochschule Dortmund, Dortmund, Germany ***M.E. Embedded Systems for Mechatronics***

September 2015 – October 2018

Visvesvaraya Technical University, India ***B.E. Instrumentation Technology***

September 2007 – May 2011

VOLUNTEERING AND OTHER PROJECTS

Autonomous Navigation & Multi-Sensor Path Planning

Focus: Embedded Sensor Integration, Real-Time Control, Autonomous Systems

- Developed control logic for an autonomous parallel parking system on a robotic vehicle prototype, using ROS-compatible embedded architecture.
- Integrated and calibrated a multi-sensor array (ultrasonic HC-SR04) for real-time obstacle detection and distance estimation; implemented sensor fusion logic for reliable environmental perception.
- Implemented closed-loop control on an Arduino-based embedded platform to execute precise, real-time path-planning manoeuvres.

Stochastic Control Simulation: Anti-lock Braking System (ABS)

Focus: Mathematical Modelling, Safety-Critical Control Systems

- Modelled vehicle braking dynamics on variable friction surfaces using MATLAB Simulink.
- Applied Kalman Filters for stochastic state estimation and noise reduction in sensor data.
- Implemented and tested non-linear Bang-Bang control logic under simulated safety-critical constraints.

Mindtree.

Languages

English - Fluent

German - B2

References

Mr. Olaf Mater – Former Manager at
Marvell Semiconductors (now acquired by
Infineon)
olaf.mater@infineon.com

Mr. Pawan Kumar – Former Manager at
Marvell Semiconductors
pawan.kumar@infineon.com